

UniTS Anomaly Ratio Sweep

ESA-Mission1 | Generated 2026-03-12 10:52

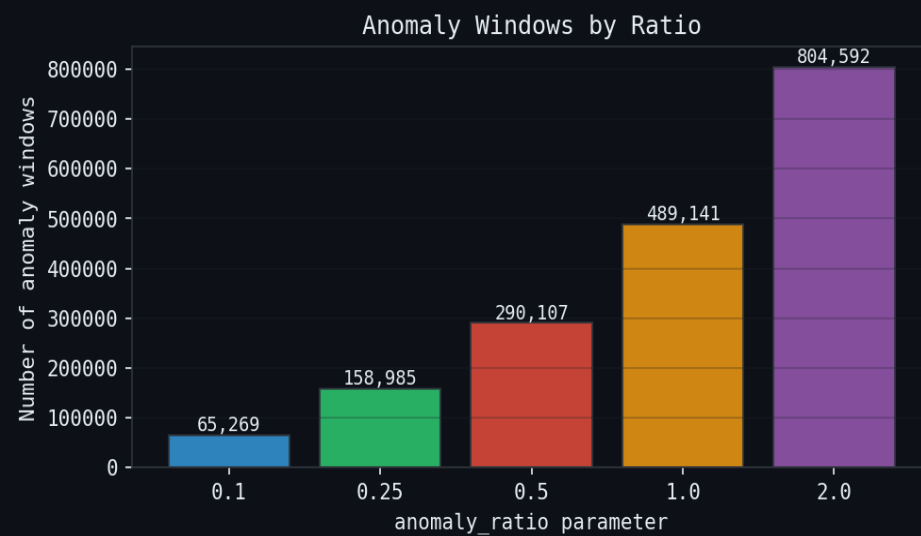
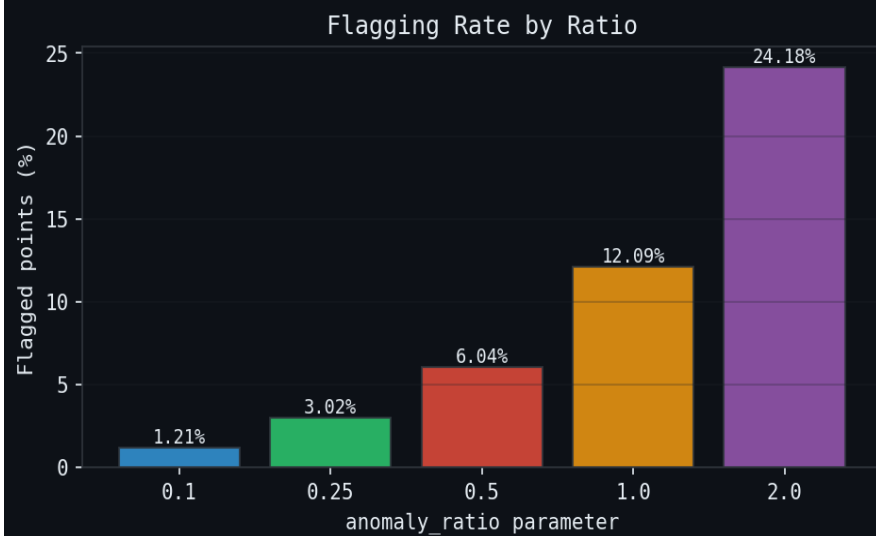
Results by Ratio

anomaly_ratio	Threshold	Flagged Points	Flag Rate	Windows	Precision	Recall	F1
0.1	0.147728	89,025	1.209%	65,269	0.9210	0.9966	0.9573
0.25	0.124461	222,563	3.022%	158,985	0.8121	0.9972	0.8952
0.5	0.106515	445,127	6.045%	290,107	0.6779	0.9973	0.8072
1.0	0.088345	890,254	12.089%	489,141	0.5098	0.9984	0.6750
2.0	0.071131	1,780,509	24.178%	804,592	0.3408	0.9984	0.5082

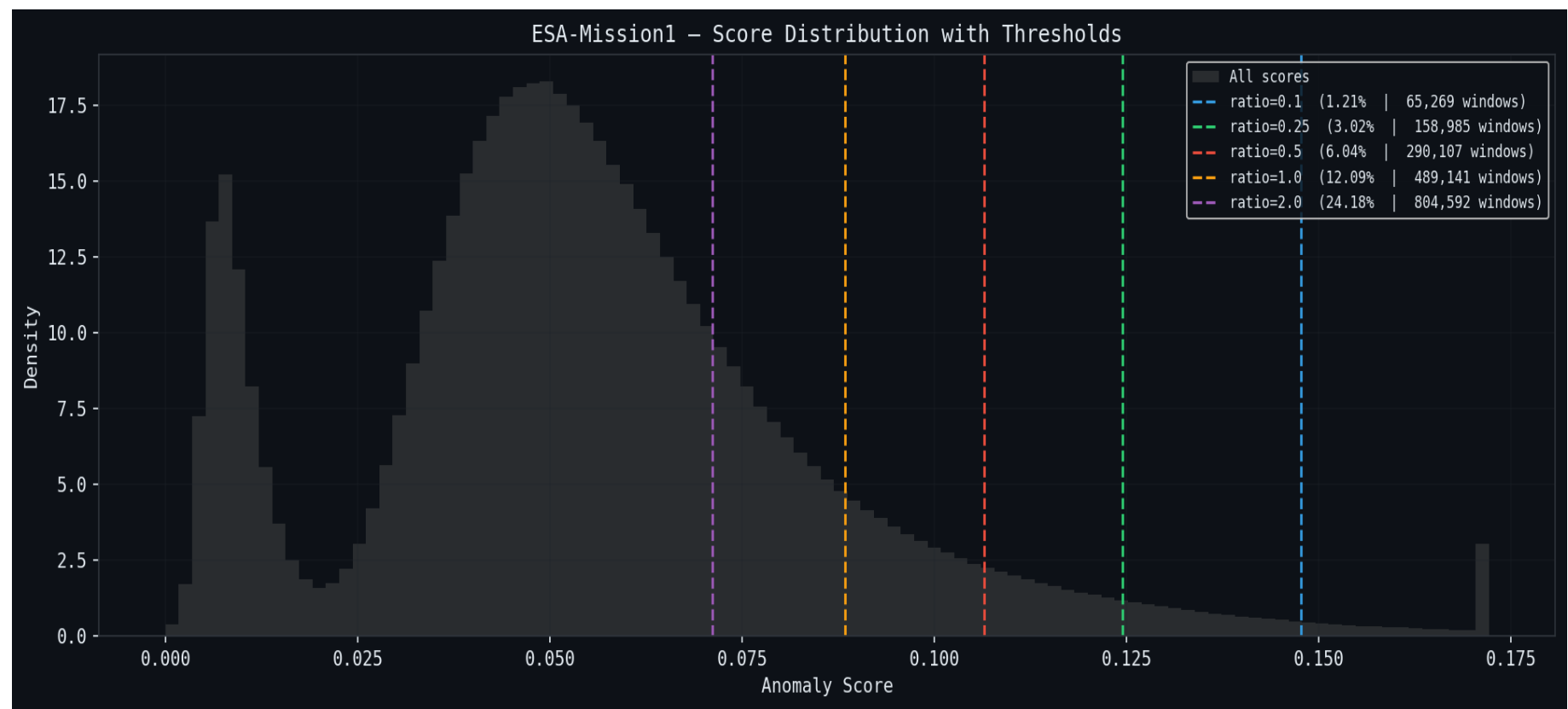
anomaly_ratio controls the detection threshold: UniTS flags the top ratio% of reconstruction error scores as anomalies. Lower values = stricter threshold = fewer flags. The threshold column shows the exact score cutoff for each ratio.

Flagging Rate & Window Count

ESA-Mission1 – Ratio Sweep Summary



Score Distribution with Thresholds



Dashed vertical lines show where each ratio sets its threshold. Choose a ratio whose threshold sits clearly in the tail of the distribution rather than in the main body of normal scores.

Daily Flagging Rate by Ratio

ESA-Mission1 – Daily Flagging Rate by Ratio

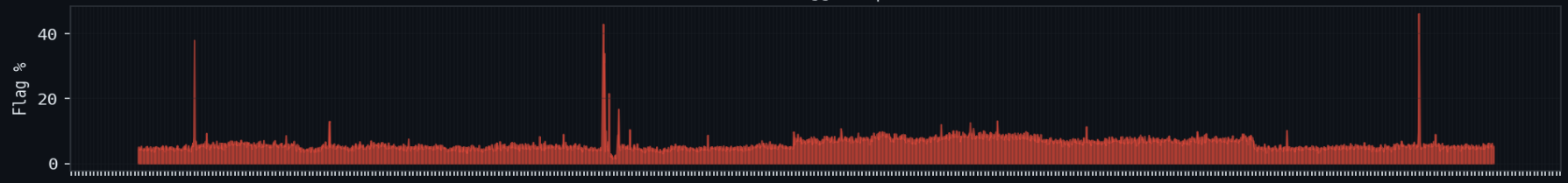
ratio=0.1 → 1.21% flagged | 65,269 windows



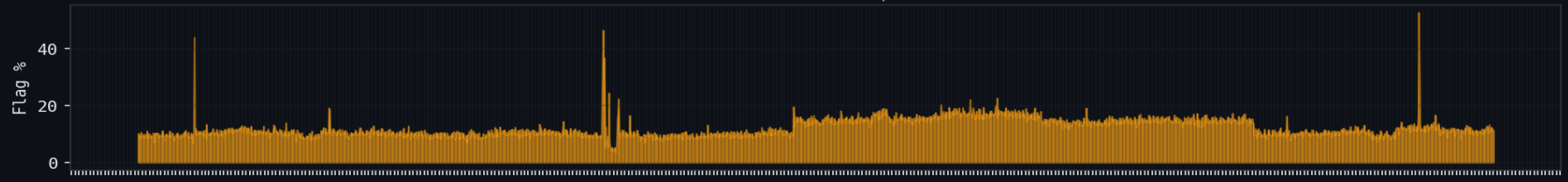
ratio=0.25 → 3.02% flagged | 158,985 windows



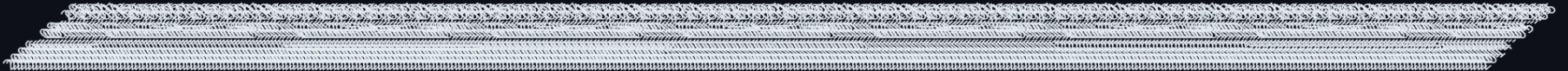
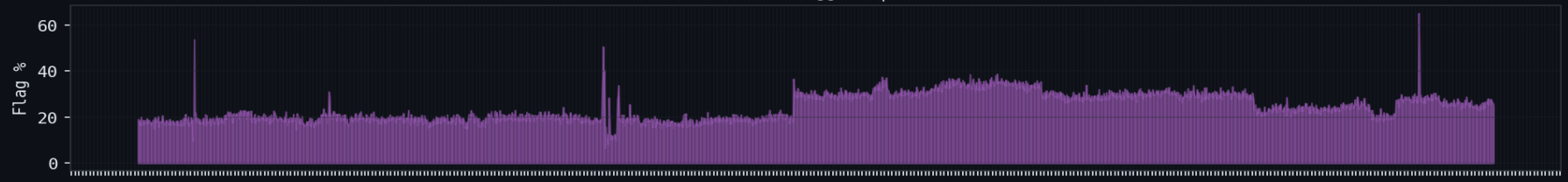
ratio=0.5 → 6.04% flagged | 290,107 windows



ratio=1.0 → 12.09% flagged | 489,141 windows



ratio=2.0 → 24.18% flagged | 804,592 windows



A realistic anomaly rate for a nominal satellite is typically 0.1–1%. Look for a ratio that shows isolated spikes rather than sustained elevated flagging.